

THE HIGH COST OF OPERATING IN THE DARK.

Managing a large-scale ride-hailing fleet is a constant battle against operational blind spots.

Without granular visibility into hourly demand curves and vehicle-level margins, fleet deployment remains entirely reactive.

The Result: Longer wait times, lower driver earnings, and millions left on the table in missed surge revenue.



Mapping the pulse of NYC mobility.

Dataset Context: Complete visibility into urban ride-hailing performance across NYC (June 1–30, 2024)

103,728

Total Bookings



\$1.6M

Total Booking Value



349,000

Total Miles Covered



16 Min

Average Trip Time



From raw logs to strategic intelligence.

Data Refinery



1. Extraction (The Raw Material)

Flat CSV files representing 103K+ disparate trip records.



2. Structuring (The Engine Room)

PostgreSQL database. Ensuring data integrity, handling missing location fields.



3. Modeling (The Brain)

Application of Machine Learning (Prophet, KMeans) to forecast and segment.



4. Illumination (Mission Control)

Power BI. Dynamic drill-throughs, custom DAX, and matrix heatmaps for executive consumption.

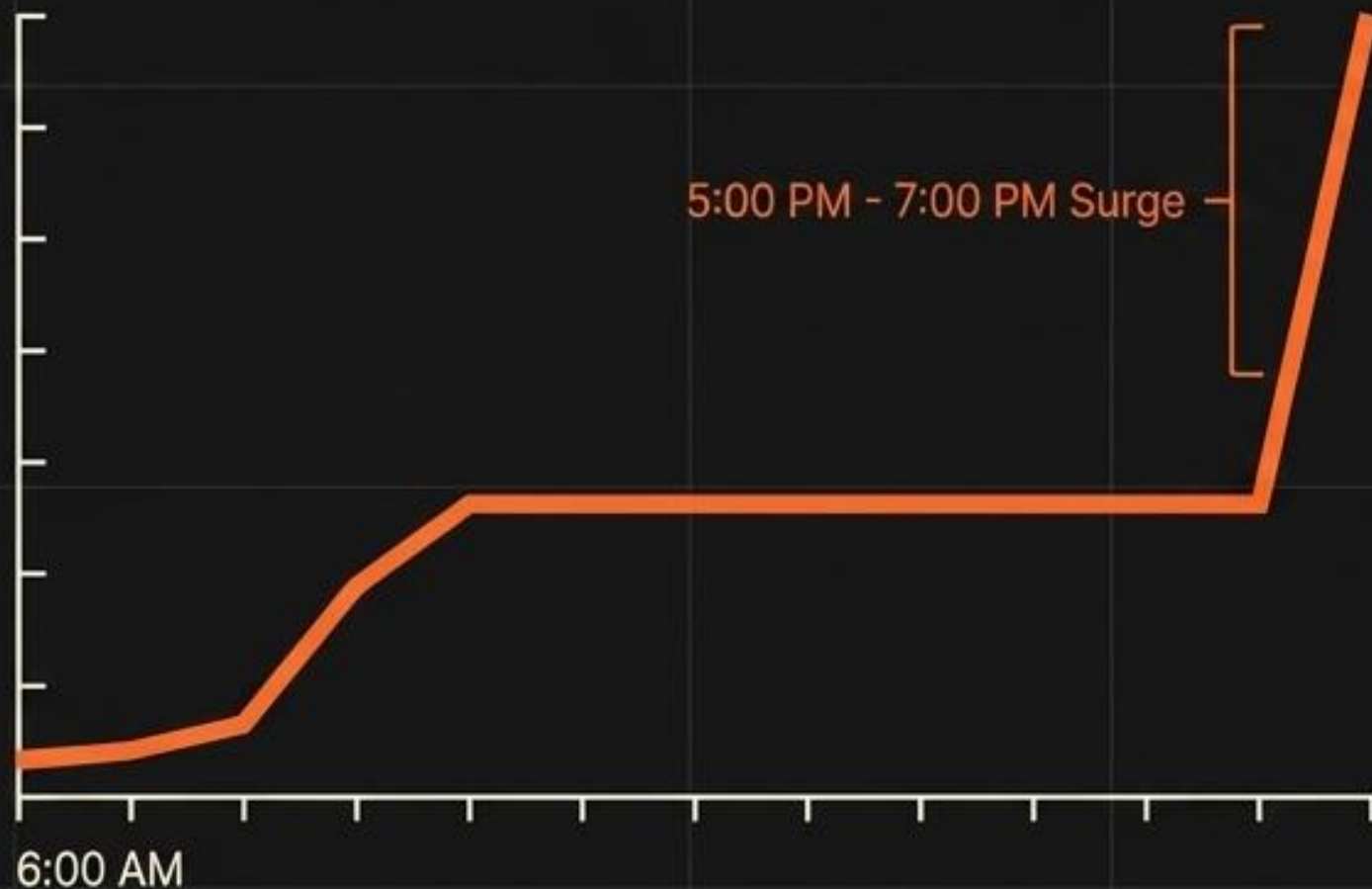
Machine Learning in reality, not just in theory.

Model	Confidence	Strategic Caveat
Demand Forecast (Prophet)		Directionally useful for 7-day seasonality. Needs rigorous resampling to hourly buckets before production deployment.
Customer Segmentation (KMeans)		4 distinct clusters identified. Requires formal elbow/silhouette validation before driving marketing campaigns.
Fare Prediction (Regression)		$R^2 = 0.989$. Unusually high. Fares are likely near-deterministic based on distance/time. A proof-of-concept, not a live pricing engine.
Payment Classification		91% accuracy is a mirage. Severe class imbalance (massive Uber Pay/Cash dominance vs. near-zero Amazon/Google Pay).

The spatial and temporal edge.

Temporal Demand Curve

Night trips dominate (65.3% volume / 67.7K bookings).
Weekends peak at \$283K (Sunday) vs \$146K (Mid-week).



Geographic Revenue Hubs

Top Revenue Hubs: LaGuardia Airport (\$74K),
Penn Station (\$63K), JFK Airport (\$62K).



The Workhorses vs. The Margin Makers



The Workhorse: UberX

- Volume: 38,744 Bookings (Market Leader)
- Revenue: \$583K
- Role: Maintains baseline fleet utilization and covers maximum distance (131.5K miles).

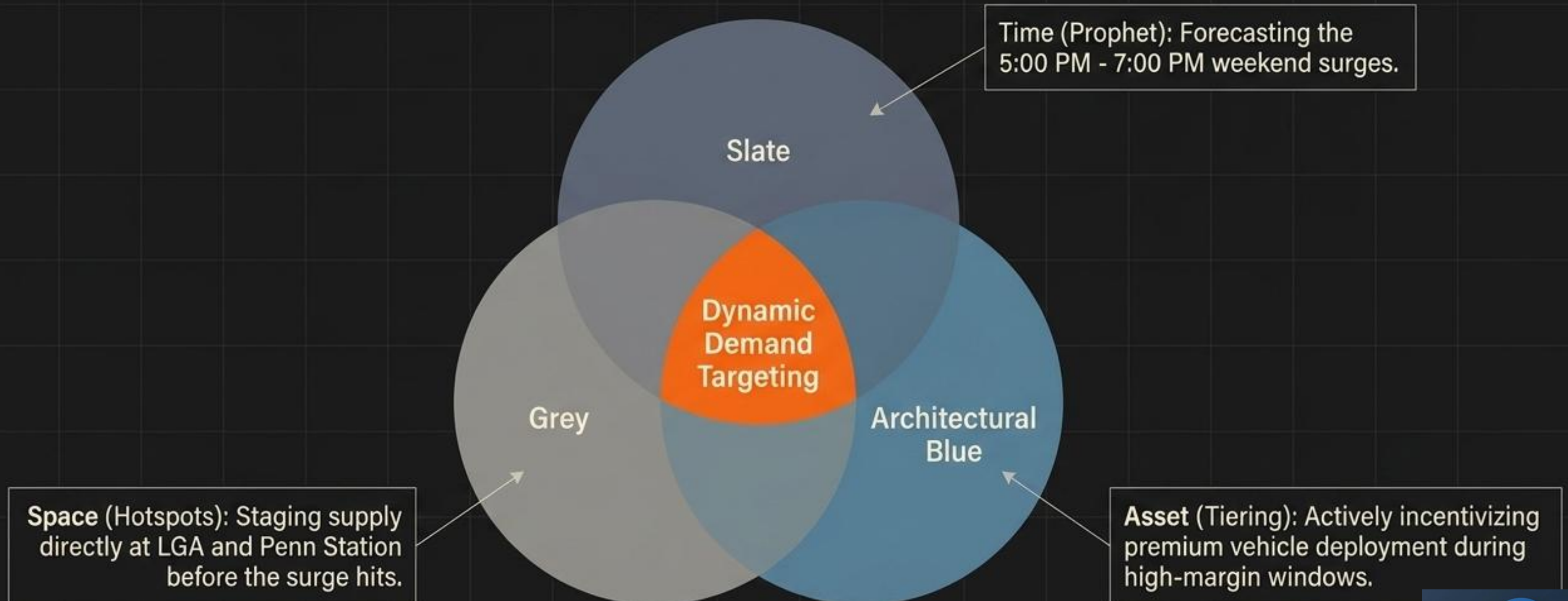


The Margin Makers: Uber Black, Comfort, & XL

- Volume: ~50,000 Bookings Combined
- Revenue: > \$750K Combined (\$250K Black + \$254K Comfort + \$249K XL)
- Role: Disproportionate revenue contributors per trip. Prime target for retention and upsell strategies.

The shift to Predictive Supply Strategy

Reactive fixed shifts are obsolete. By synthesizing our three data pillars, fleet deployment becomes an active weapon.



Erasing blind spots with Mission Control.

Total Bookings:
88,744

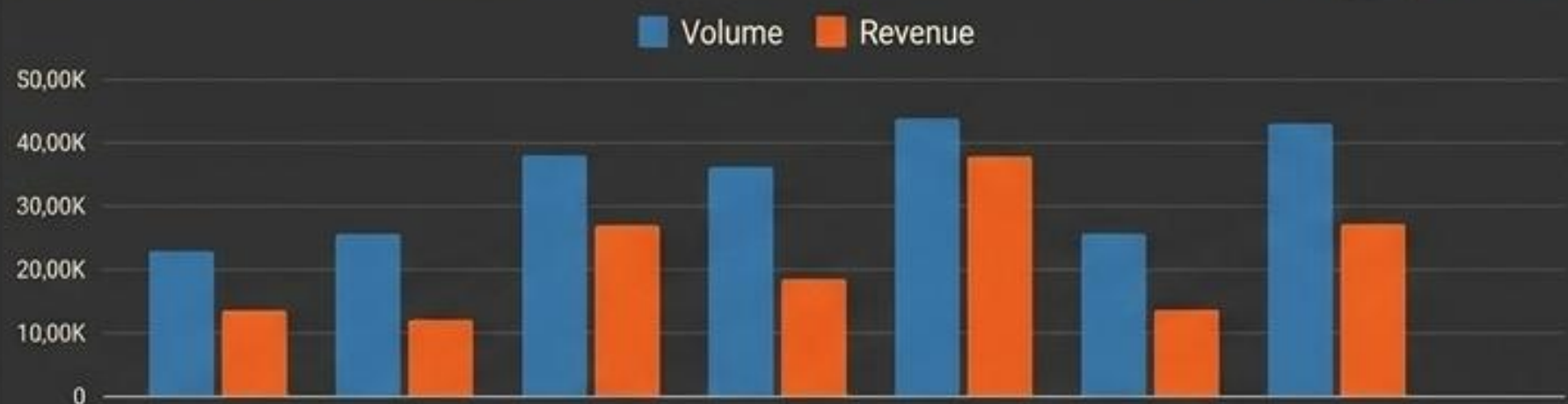
Total Value:
\$1.3M

Total Distance:
385.5K miles

Executive KPIs: Instant visibility into Total Bookings, Value, and Distance.

Volume vs. Revenue Performance

View: **Volume** ☐ **Revenue**



Dynamic Visual Toggles: Switch seamlessly between volume and revenue views.

Trip ID	Surge Fee	Passenger Count	Status	Amount
T-ID-451	\$5.50	2	Completed	\$32.50
T-ID-452	\$5.50	2	Completed	\$32.50
T-ID-453	\$5.50	2	Completed	\$32.50
T-ID-454	\$5.50	2	Completed	\$32.50

Row-Level Drill-Through: Granular transaction visibility down to the specific Trip ID, Surge Fee, and Passenger Count



Stuck in Visualization ?

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